NAME: Aryan Singh STD.: ________ SEC.: ________ ROLL NO.: ________ SUB.: ________

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
		<u>Index</u>		
2	12-Aug-25	full adder and subtractor verilog code		12/8/25
3	26-Aug-25	Decoder using test Bench		26/8/25
4	26-Aug-25	Multiplexer And Demultiplexer code in verilog		26/8/25

(*) **Aim** :- To design stimulate and verify the Magnification functionally of various fundamental combinational logic circuits using verilog HDL.

(*) **Theory** :-

This experiment covers the Design And verification of standard combinational circuits and fundamental concepts of the verilog language.

(*) **Tools required** :-

(1) **Model Sim** :- This is an HDL simulation tool used for verifying the functionality of digital circuits.

(2) **Verilog HDL** :- This is the language used to describe the digital circuits and their best benches.

Date
12-Aug-25

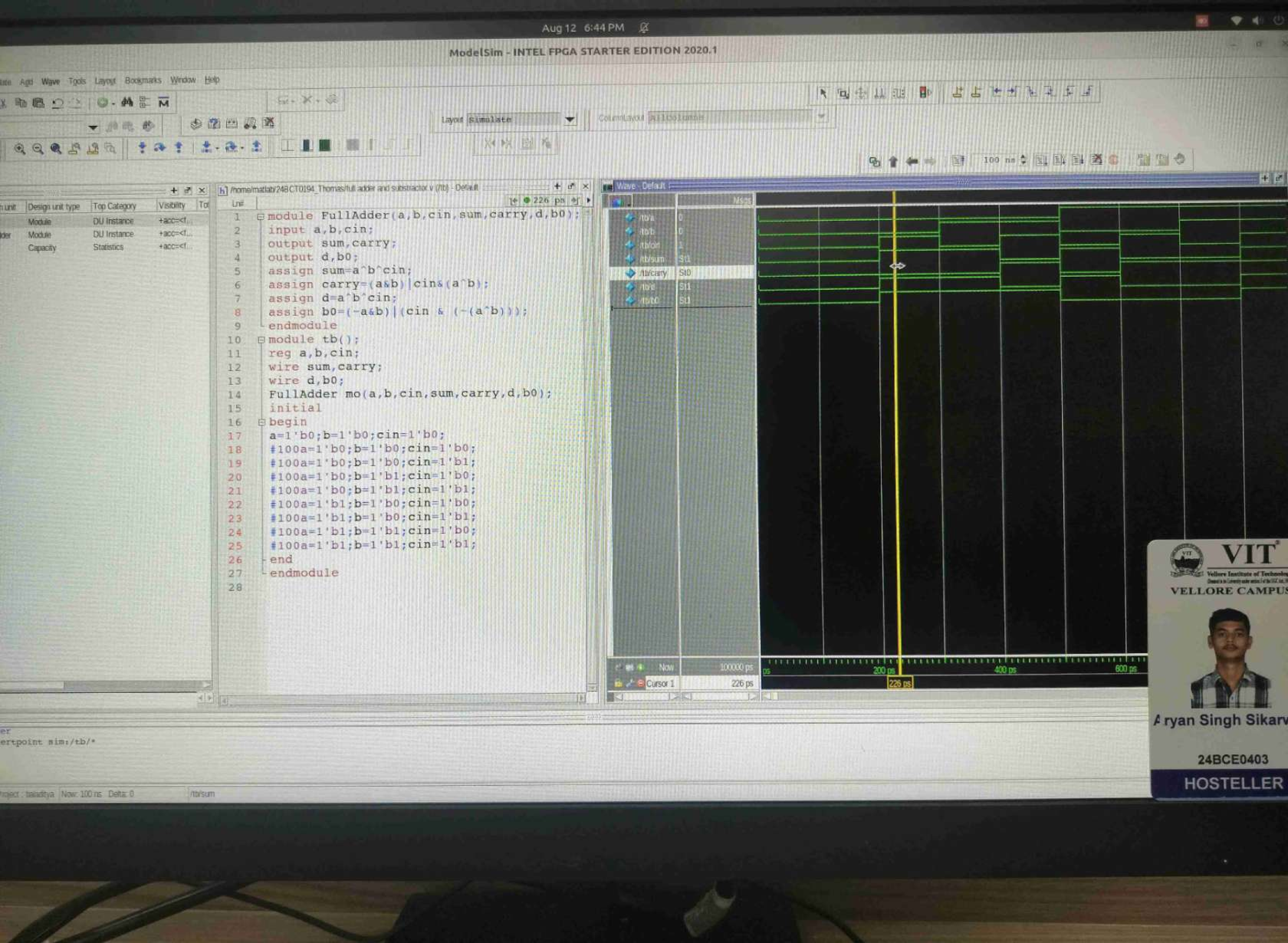
{ Assesment - 2 }

full Adder

```
module full_adder (a, b, cin, sum, carry, d, bo);
input a, b, cin;
output sum, caury;
output d, bo;
assign sum = a ^ b ^ cin;
assign carry = (a & b) | cin & (a ^ b);
assign d = a ^ b ^ cin;
assign bo = ((~a) & b) | ((a ^ b) & (~cin));
endmodule
module tb();
reg a, b, cin;
wire sum, carry;
wire d, bo;
full_Adder m0 (a, b, cin, sum, carry, d, bo);
initial
begin
a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
end
endmodule
```

o/p verified
12/8/25
24BCE 0403

```
module Full_Adder(a,b,cin,sum,carry,d,b0);
input a,b,cin;
output sum,carry;
output d,b0;
assign sum=a^b^cin;
assign carry=(a&b)|cin&(a^b);
assign d=a^b^cin;
assign b0=((~a)&b)|((a^b)&(~cin));
endmodule
module tb();
reg a,b,cin;
wire sum,carry;
wire d,b0;
Full_Adder mo(a,b,cin,sum,carry,d,b0);
initial
begin
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
end
endmodule
```


Aug 12 6:44 PM
ModelSim - INTEL FPGA STARTER EDITION 2020.1
Wave - Default
module FullAdder(a,b,cin,sum,carry,d,b0);
input a,b,cin;
output sum,carry;
output d,b0;
assign sum=a^b^cin;
assign carry=(a&b)|cin&(a^b);
assign d=a^b^cin;
assign b0=(~a&b)|(cin & (~(a^b)));
endmodule
module tb();
reg a,b,cin;
wire sum,carry;
wire d,b0;
FullAdder mo(a,b,cin,sum,carry,d,b0);
initial
begin
a=1'b0;b=1'b0;cin=1'b0;
#100a=1'b0;b=1'b0;cin=1'b0;
#100a=1'b0;b=1'b0;cin=1'b1;
#100a=1'b0;b=1'b1;cin=1'b0;
#100a=1'b0;b=1'b1;cin=1'b1;
#100a=1'b1;b=1'b0;cin=1'b0;
#100a=1'b1;b=1'b0;cin=1'b1;
#100a=1'b1;b=1'b1;cin=1'b0;
#100a=1'b1;b=1'b1;cin=1'b1;
end
endmodule
/tb/a
/tb/b
/tb/cin
/tb/sum
/tb/carry
/tb/d
/tb/b0
Now 100000 ps
Cursor 1 226 ps
200 ps
400 ps
600 ps
VIT
VELLORE CAMPUS
24BCE0403
HOSTELLER

Page

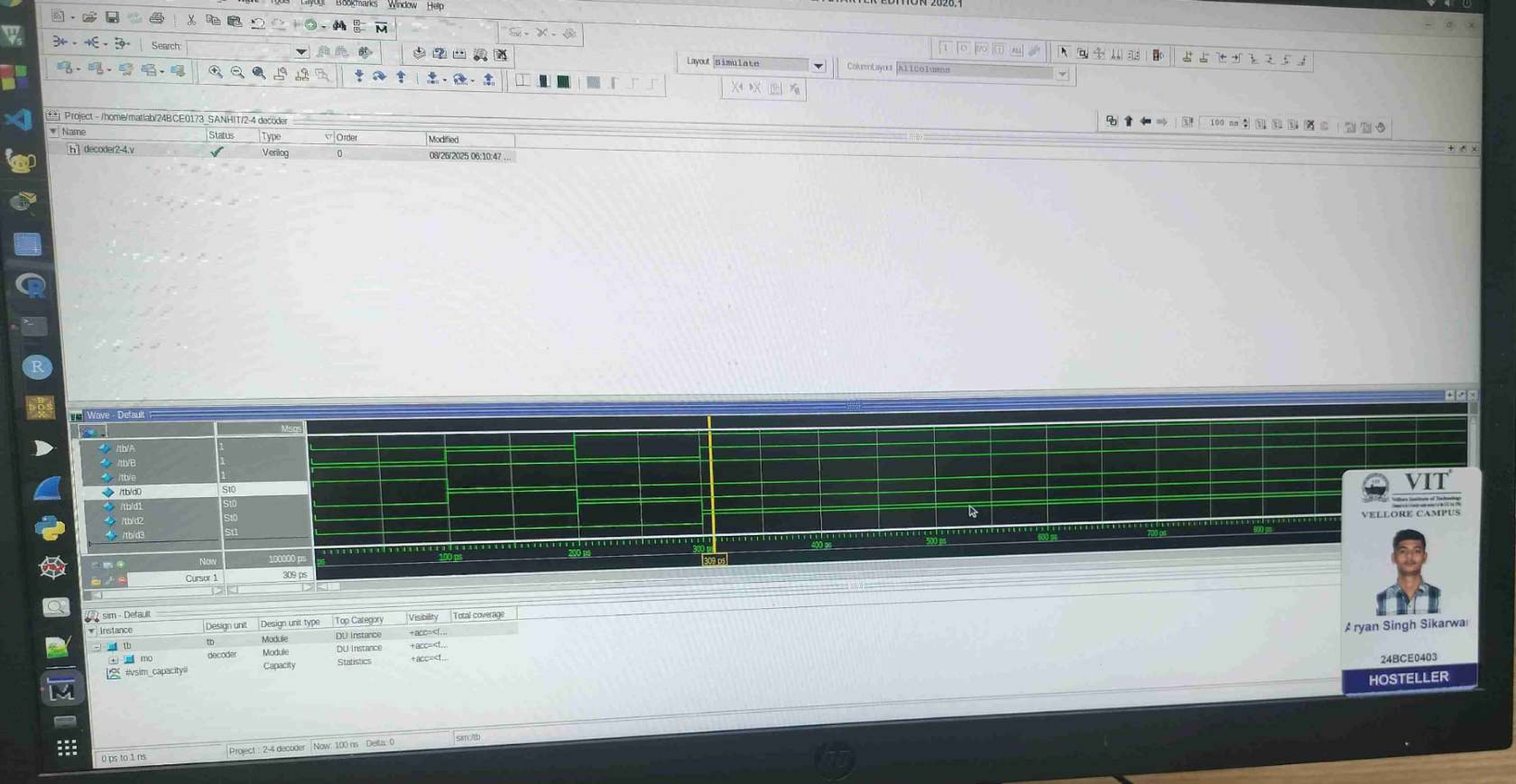
$\left\{\begin{array}{l} T \to 0 \\ F = 1 \end{array}\right\}$

A	B	C	D	Bo	
T	T	T	T	T	
T	T	F	F	F	
T	F	T	F	F	
T	F	F	T	F	
F	T	T	F	T	
F	T	F	T	T	
F	F	T	T	T	
F	F	F	F	F	

#2 Decoder Test Bench code :-

```
module decoder (A,B,e,d0,d1,d2,d3);
input A,B;
output d0,d1,d2,d3;
assign d0 = ~A & ~B & e;
assign d1 = ~A & B & e;
assign d2 = A & ~B & e;
assign d3 = A & B & e;
endmodule tb();
reg A,B,e;
wire d0,d1,d2,d3;
decoder dec (A,B,e,d0,d1,d2,d3);
initial
begin
A=1'b0; B=1'b0; e=1'b1;
#100 A=1'b0; B=1'b0; e=1'b1;
#100 A=1'b1; B=1'b0; e=1'b1;
#100 A=1'b1; B=1'b1; e=1'b1;
end
endmodule
```

ole verified
26/8/25
24 BCE0403



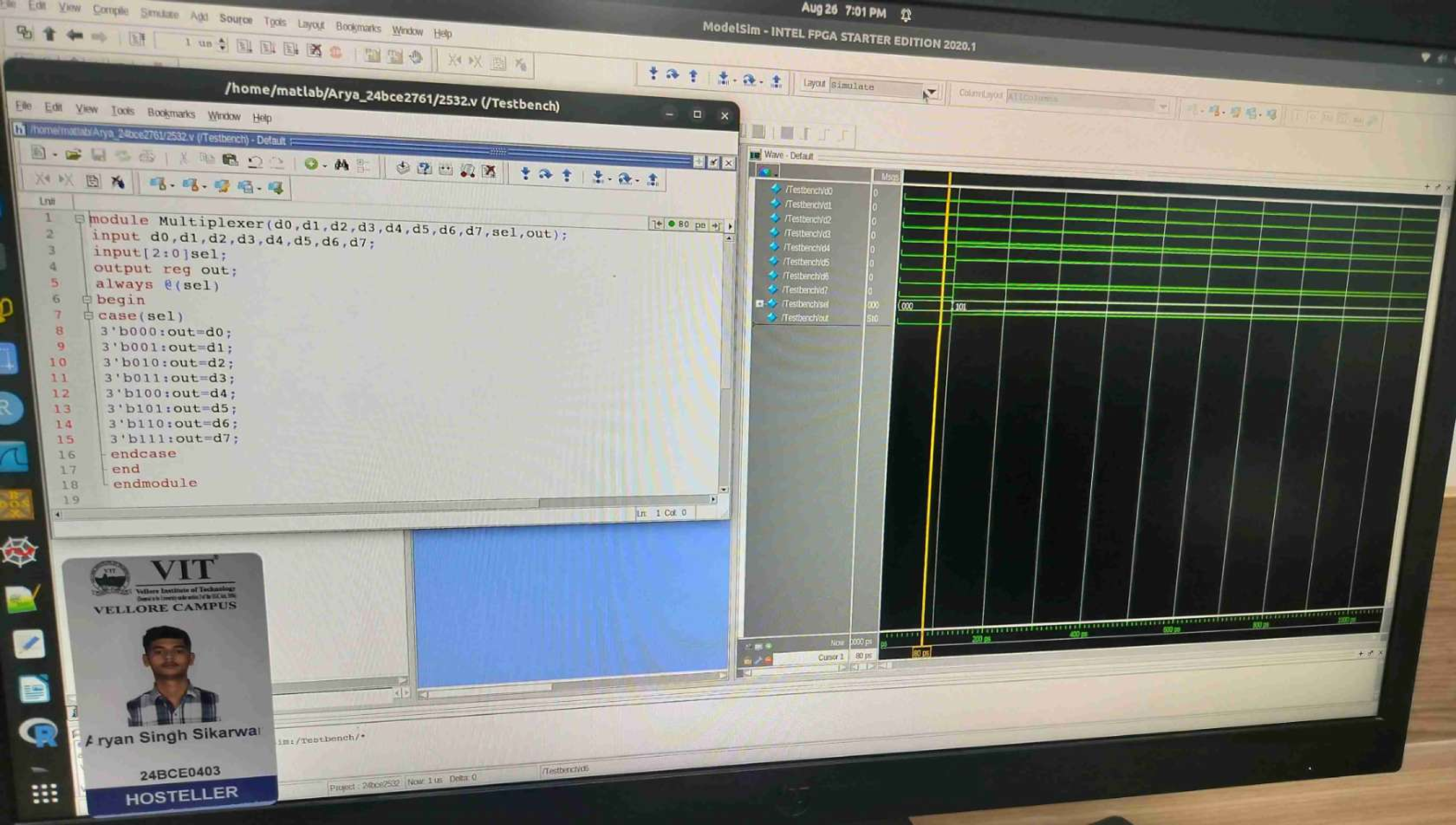
Bookmarks
Window
Help
Search:
Layout
simulate
ColumnLayout
AllColumns
Project - /home/matlab/24BCE0173_SANHIT/2-4 decoder
Name
Status
Type
Order
Modified
decoder2-4.v
Verilog
0
08/26/2025 06:10:47 ...
Wave - Default
Msgs
/tb/A
/tb/B
/tb/e
/tb/d0
/tb/d1
/tb/d2
/tb/d3
1
1
1
St0
St0
St0
St1
Now
100000 ps
Cursor 1
309 ps
100 ps
200 ps
300 ps
400 ps
500 ps
600 ps
700 ps
309 ps
sim - Default
Instance
Design unit
Design unit type
Top Category
Visibility
Total coverage
tb
mo
#vsim_capacity#
tb
decoder
Module
Module
Capacity
DU Instance
DU Instance
Statistics
+acc=<f...
+acc=<f...
+acc=<f...
0 ps to 1 ns
Project : 2-4 decoder
Now: 100 ns
Delta: 0
sim:/tb
VIT
VELLORE CAMPUS
Aryan Singh Sikarwar
24BCE0403
HOSTELLER

Multiplexer Code :-

```
module Multiplexer (d0, d1, d2, d3, d4, d5, d6
, d7, sel, out) ;
input d0, d1, d2, d3, d4, d5, d6, d7;
input [2:0] sel ;
output reg out ;
always @ (sel) .
begin
case (sel) .
3'b000 : out d0 ;
3'b001 : out d1 ;
3'b010 : dot d2 ;
3'b011 : out d3 ;
3'b100 : aot d4 ;
3'b101 : out d5 ;
3'b110 : out d6 ;
3'b111 : out d7 ;
end
endmodule
```

24BCE0403

```
module Testbench ;
reg d0, d1, d2, d3, d4, d5, d6, d7;
reg [2:0] sel ;
wire out ;
multiplexer uut (.d0(d0), .d1(d1), .d2(d2)
, .d3(d3), .d4(d4), .d5(d5), .d6(d6)
, .d7(d7), .sel(sel), .(out)out));
initial begin
d0=0 ; d1=0 ; d2=0 ; d3=0 ; d4=0;
d5=0 ; d6=0 ; d7=0 ; sel=0 ;
# 100 ;
d0=0 ; d1=0 ; d2=0 ; d3=0 ;
d4=1 ; d5=1 ; d6=0 ; d7=1 ;
```


Aug 26 7:01 PM
ModelSim - INTEL FPGA STARTER EDITION 2020.1
/home/matlab/Arya_24bce2761/2532.v (/Testbench)
module Multiplexer(d0,d1,d2,d3,d4,d5,d6,d7,sel,out);
input d0,d1,d2,d3,d4,d5,d6,d7;
input[2:0]sel;
output reg out;
always @(sel)
begin
case(sel)
3'b000:out=d0;
3'b001:out=d1;
3'b010:out=d2;
3'b011:out=d3;
3'b100:out=d4;
3'b101:out=d5;
3'b110:out=d6;
3'b111:out=d7;
endcase
end
endmodule
Wave - Default
/Testbench/d0
/Testbench/d1
/Testbench/d2
/Testbench/d3
/Testbench/d4
/Testbench/d5
/Testbench/d6
/Testbench/d7
/Testbench/sel
/Testbench/out
000
101
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Aryan Singh Sikarwar
24BCE0403
HOSTELLER

Demultiplexer Code :-

```
module demultiplexer (a,x,y,z,d0,d1,d2,
  d3,d4,d5,d6,d7);
input a,x,y,z;
output d0,d1,d2,d3,d4,d5,d6,d7;
assign d0 = (a&~x&~y&~z);
d1 = (a&~x&~y&z);
d2 = (a&~x&y&~z);
d3 = (a&~x&y&z);
d4 = (a&x&y&~z);
d5 = (a&x&y&z);
d6 = (a&x&y&~z);
d7 = (a&x&y&z);
endmodule

module Testbench_demux;
reg a,x,y,z;
wire d0,d1;d2,d3,d4,d5,d6,d7;
demultiplexer out(.a(a), x(x), .y(y),
.z(z), .d0(d0), .d1(d1), .d2(d2), .d3(d3),
.d4(d4), .d5(d5), .d6(d6), .d7(d7));
initial
begin
a=0; x=0; y=0; z=0;
#100 a=1; x=0; y=1; z=0;
#100 a=1; x=1; y=0; z=0;
#100 a=1; x=1; y=1; z=1;
end
endmodule
```

o/p Verified
26/08/25
24BCE0403

Activities   WindowObj   Aug 26 6:49 PM

ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help

Layout Simulate   ColumnLayout AllColumns

sim - Default

Instance	Design unit	Design unit type	Top Category	Visibility	Total coverage
tb	tb	Module	DU Instance	+acc=<f...	
uut	de_mux	Module	DU Instance	+acc=<f...	
#vsim_capacity#		Capacity	Statistics	+acc=<f...	

/home/matlab/zayn_mux.v (/tb) - Default

```verilog
module de_mux(a,x,y,z,d0,d1,d2,d3,d4,d5,d6,d7);
input a,x,y,z;
output d0,d1,d2,d3,d4,d5,d6,d7;
assign d0 = (a& ~x& ~y&~z),
d1 = (a&~x&~y&z),
d2 = (a&~x&y&~z),
d3 = (a&~x&y&z),
d4 = (a&x&y&~z),
d5 = (a&x&y&z),
d6 = (a&x&y&~z),
d7 = (a&x&y&z);
endmodule
module tb;
reg a,x,y,z;
wire d0,d1,d2,d3,d4,d5,d6,d7;
de_mux uut(.a(a),.x(x),.y(y),.z(z),.d0(d0),.d1(d1),.d2(d2),.d3(d3),.d4(d4),.d5(d5),.d6(d6),.d7(d7));
initial begin
a=0;x=0;y=0;z=0;
#100 a=1;x=0;y=1;z=0;
#100 a=1;x=1;y=0;z=0;
#100 a=1;x=1;y=1;z=1;
end
endmodule
```

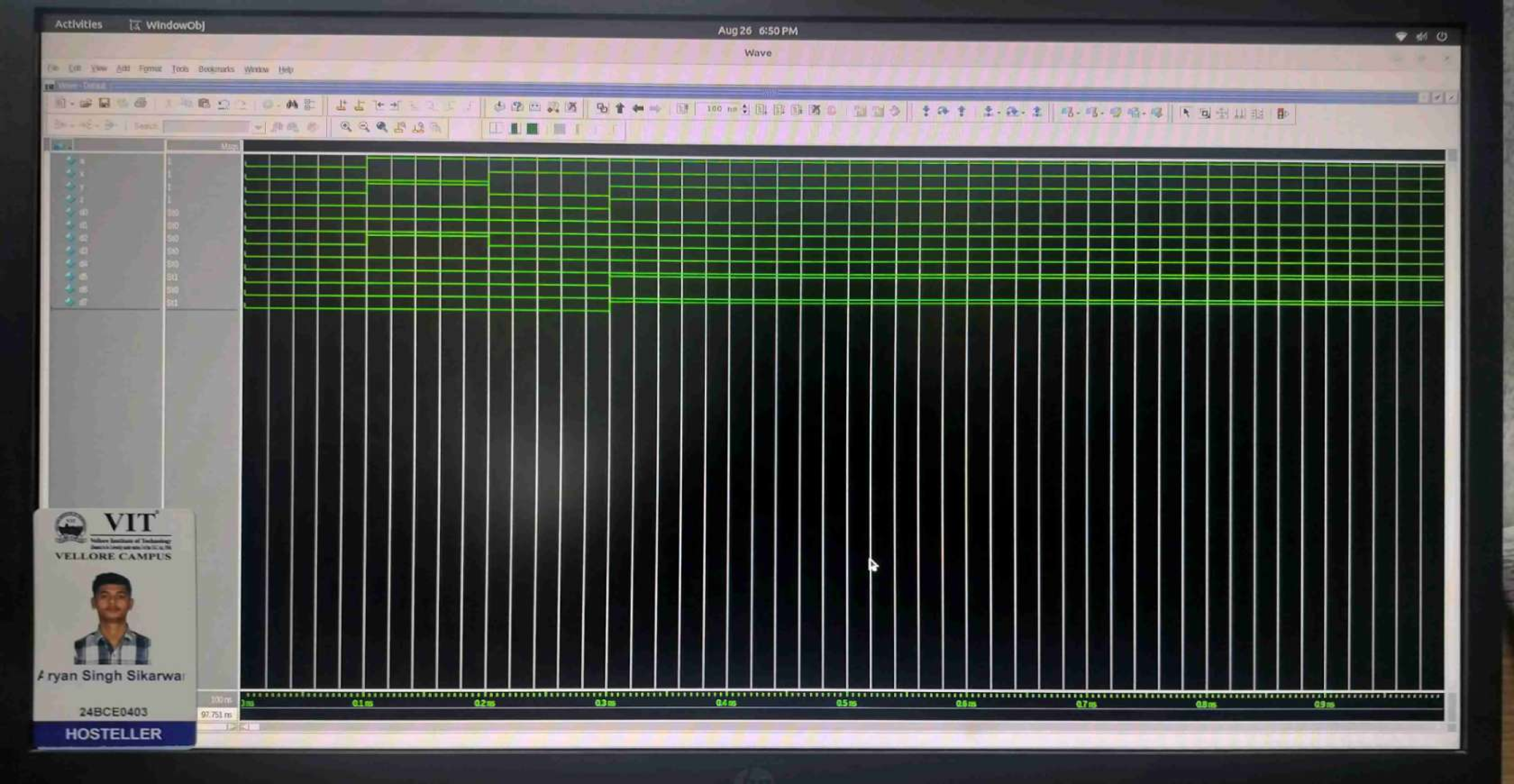

```
0 (50 ps) -r 100
d with 3 errors.
d with 2 errors.
d with 5 errors.
uccessful.
d with 5 errors.
uccessful.
uccessful.

6,2025, Elapsed time: 0:24:58

26,2025

: sim:/tb/*
```

mux   Now: 100 ns Delta: 0   sim:/tb


Activities
WindowObj
Aug 26 6:50 PM
Wave
File Edit View Add Format Tools Bookmarks Window Help
VIT
VELLORE CAMPUS
ryan Singh Sikarwa
24BCE0403
HOSTELLER

ModelSim - INTEL FPGA STARTER EDITION 2020.1

/home/matlab/debu.v (/tb) - Default

```
module decorder(A,B,e,d0,d1,d2,d3);
input A,B,e;
output d0,d1,d2,d3;
assign d0 = ~A & ~B & e;
assign d1 = ~A & B & e;
assign d2 = A & ~B & e;
assign d3 = A & B & e;
endmodule

module tb();
reg A,B,e;
wire d0,d1,d2,d3;
decorder dec(A,B,e,d0,d1,d2,d3);
initial
begin
A=1'b0;B=1'b0;e=1'b1;
#100 A = 1'b0;B = 1'b1;e=1'b1;
#100 A = 1'b1;B = 1'b0;e=1'b1;
#100 A = 1'b1;B = 1'b1;e=1'b1;
end
endmodule
```

Project : debu | Now: 100 ns Delta: 0 | sim:/tb

Answer the following

* What do you mean by test Bench in verilog?

Ans It creates a copy of the module you want to test and places it inside the virtual lab. It seprates piece of a code you write to check if your main hardware design.

* What is Difference between initial and always?

Ans The initial block runs only piece at the beginning of a simulation typically for setting up a test. The always block runs repeatedly and continuosly, modeling the behaviour of real hardware always active.

* What is installation in verilog?

Ans Instantiation is the process of creating a usable copy of one module inside another. Its the fundamental method for building complex, hierarchical circuits from smaller, reusable components.